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Letter

High-Performance Atomic-Layer-Deposited Dual-Gate InGaO Thin-Film Transistors

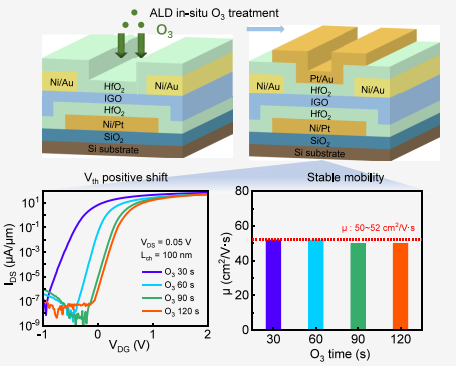
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ABSTRACT: Amorphous oxide semiconductors (AOSs) have attracted considerable attention because of their high carrier density, low thermal budget, and large bandgap. However, the high electron density in AOSs hinders their ability to turn off effectively, resulting in a trade-off between the threshold voltage (V_{th}) and mobility (μ). In this work, we report high-performance dual-gate (DG) indium gallium oxide (IGO) TFTs utilizing localized O_3 treatment to effectively passivate the oxygen vacancies (V_o) in the channel region of IGO TFTs, thereby achieving a positive V_{th} and high mobility. The 100 nm short-channel length (L_{ch}) enhancement-mode IGO DG TFT exhibits an ideal subthreshold slope (SS) of 63 mV/dec, a maximum drain current I_{DS} of 1.36 mA/μm, and a record high transconductance (g_m) of 1008 μS/μm. This study demonstrates a novel method to overcome the trade-off between V_{th} and μ, showing that IGO DG-TFTs are promising transistors for enabling high-performance monolithic three-dimensional (M3D) integrated circuits.



KEYWORDS: *indium−gallium oxide, dual-gate TFTs, O_3 treatment, mobility*

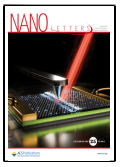
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In recent years, AOSs have significantly advanced storage technologies, display innovations, and flexible electronics due to their exceptional properties, including high electron mobility, wide bandgap (>3 eV), reduced processing temperatures (<400 °C), and excellent large-area uniformity.[1−6] In these applications, ultrathin films at the nanometer scale are required to be uniformly deposited on 3D structures, necessitating the use of atomic layer deposition (ALD). ALD can uniformly deposit thin films in the nanoscale thickness class onto intricate 3D structures, and it offers more advantages than sputtering deposition of thin films in application scenarios such monolithic 3D integration and 3D-structured memory devices. Compared with conventional sputtering, ALD-derived a-IGZO films exhibit a higher packing density, mobility, and reliability.[7−10] Among these AOSs, amorphous In_2O_3, a key AOS material of ALD, has garnered considerable attention. The Fermi level of In_2O_3 is deeply embedded within the conduction band, resulting in a high electron concentration.[11] This characteristic facilitates exceptional electron mobility, exceeding 50 $cm^2/(V{\cdot}s)$. However, its high carrier concentration leads to a significantly negative threshold voltage, which is detrimental to meeting the power consumption requirements of devices.[12] To address this issue, the ternary compound IGO has emerged as a promising alternative. The stronger Ga−O bonds in IGO effectively suppress oxygen vacancies, thereby reducing the electron concentration.[13−15] This modification enables IGO to turn off effectively while preserving the inherent high mobility of In_2O_3, making it a highly promising material for future advancements.

Although IGO-based devices can effectively suppress the source-drain current, the increasing demand for enhancement-mode transistors in logic storage circuits has imposed more stringent requirements on AOS TFTs. To further increase the V_{th} of AOS, several reports have demonstrated that annealing treatments can mitigate oxygen vacancies in AOS materials,[16−18] thereby reducing the carrier concentration and increasing the V_{th}. Although O_2 annealing mitigates the issue of a negative threshold voltage, it inadvertently leads to a reduction in the defect density within the amorphous oxide channel layer, resulting in increased contact resistance and a significant decline in the intrinsic electron mobility, which in turn degrades the device's on-state current (I_{on}). This phenomenon underscores the inherent "trade-off" between V_{th} and μ in AOS materials.

To address this trade-off, we propose a localized O_3 passivation technique, which selectively passivates defects in the channel region while maintaining high carrier density in the contact regions. By optimizing the O_3 treatment time, we achieved a normally off operation in ALD-grown ultrathin IGO-based DG-TFTs. In particular, the 100 nm DG-TFT exhibits not only an ideal SS of 63 mV/dec and a positive V_{th}

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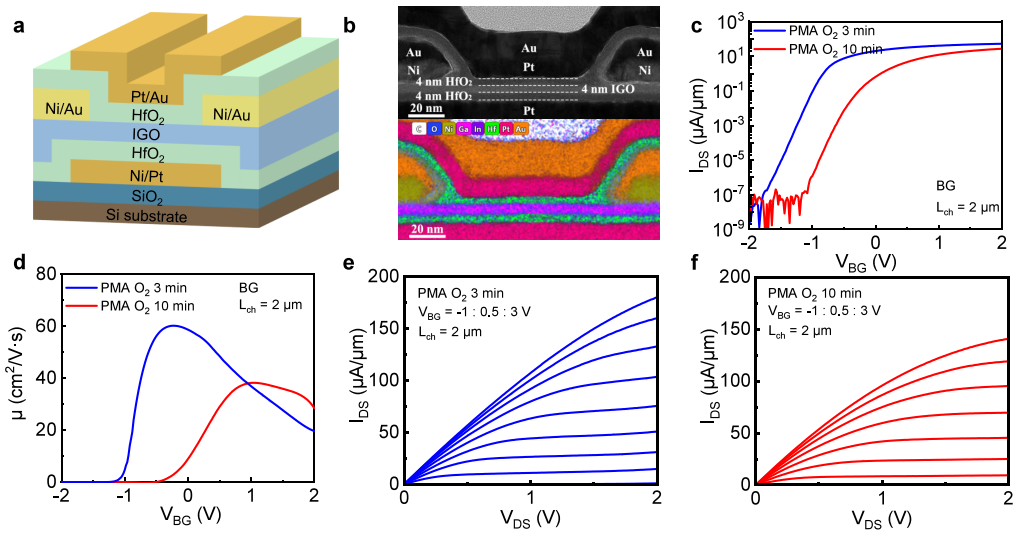


Figure 1. (a) 3D schematic of an IGO DG transistor structure. (b) TEM and EDS characterization of IGO DG transistors with 4 nm channels and dielectric layers. (c) A comparison of the transfer characteristics for L_{ch} = 2 μm under PMA durations of 3 and 10 min indicates that 10 min of annealing significantly improves V_{th}. (d) A comparison of mobility under PMA durations of 3 and 10 min. Output characteristic curves for PMA durations of (e) 3 min and (f) 10 min. The I_{DS} significantly decreases after a 10 min PMA treatment.

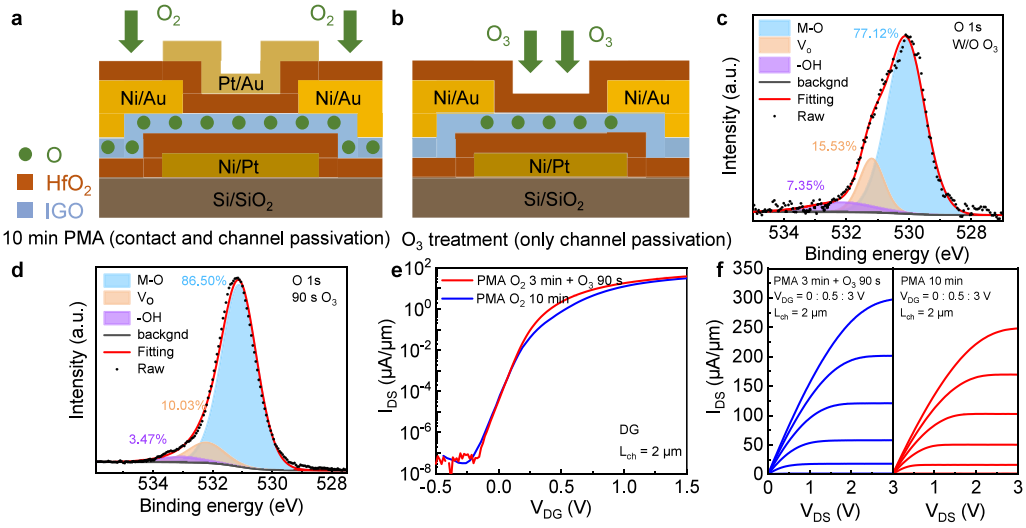


Figure 2. Schematic of (a) full passivation of oxygen vacancies (V_o) in the contact and channel regions by long-term O_2 PMA and (b) passivation of the V_o in the channel region by short-term O_3 treatment. A long O_2 PMA results in a positive V_{th} but a large R_c. Short-term O_3 treatment has a positive V_{th} and small R_c because of high V_o in the contact region. O 1s spectra of IGO films (c) before and (d) after 90 s of O_3 treatment, suggesting the reduction of V_o by O_3 treatment. (e) Comparison of the transfer characteristics between O_3 + 3 min PMA and 10 min PMA treatment alone. O_3 treatment clearly increases V_{th} while still resulting in a high I_{DS}. (f) Comparison of the output characteristics between O_3 + PMA treatment and PMA treatment alone.

of 0.15 V but also a record high g_m of 1008 μS/μm at V_{DS} = 1 V and an ultralow contact resistance (R_c) of 0.23 kΩ·μm, which indicates state-of-the-art performance in ALD ultrathin (<5 nm) DG-TFTs.

The IGO DG TFTs were meticulously designed and fabricated according to the configuration depicted in Figure 1a. The fabrication started with a substrate consisting of a 100 nm thick SiO_2 layer on silicon. Initially, the SiO_2 surface was subjected to standard RCA cleaning to eliminate surface contaminants.[19] Subsequently, electron beam lithography (EBL) was utilized to define the pattern, followed by electron beam evaporation (EBE) to deposit a 3 nm layer of Ni and a 15 nm layer of Pt, forming the back-gate (BG) electrode. The high work function of Pt was specifically chosen to ensure a positive threshold voltage for the device.[20] A 4 nm thick layer of HfO_2 was then deposited as the back-gate dielectric via ALD at 250 °C. Next, a 4 nm IGO channel layer was deposited by ALD at 225 °C. The Atomic Force Microscope (AFM) image of the material can be found in Figure S1 in the Supporting Information. Trimethylgallium (TMGa) and O_3 were employed as the precursors for Ga_2O_3, whereas trimethylindium (TMIn) and H_2O were used for In_2O_3 deposition. To achieve optimal performance and high reliability, the ALD cycle ratio of In_2O_3 to Ga_2O_3 in the IGO channel was maintained at 1:1. Channel isolation was subsequently achieved through wet etching using hydrochloric acid. The source and drain electrodes, consisting of 15 nm Ni and 15 nm Au, were defined using EBL and deposited via EBE. Following

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Nano Lett. 2025, 25, 8541−8546

the standard lift-off process, a 4 nm layer of HfO_2 was deposited by O_3-based ALD as the top-gate (TG) dielectric at 250 °C. To selectively passivate defects within the IGO channel, an O_3 treatment was applied. This approach avoids defect passivation in the source/drain regions, as observed with postmetallization annealing (PMA), thereby preventing an increase in contact resistance. Finally, a Pt/Au TG electrode (5 nm/25 nm) was deposited using EBE. The channel length (L_{ch}) of the DG TFTs ranged from 100 nm to 2 μm, with a constant channel width of 1 μm. A scanning electron microscopy (SEM) image of the device with an L_{ch} of 100 nm can be found in Figure S1 of the Supporting Information. The XPS images of In 3d and Ga 2p are shown in Figure S2 of the Supporting Information. Transmission electron microscopy (TEM) images of the cross sections of the IGO DG TFTs, as shown in Figure 1b, clearly reveal a well-defined interface between the channel and the gate dielectrics. The thicknesses of the IGO channel, BG dielectric, and TG dielectric were all precisely maintained at 4 nm. Consequently, the TG dielectric and BG dielectric exhibit equivalent EOT values. In addition, the gate leakage current is plotted in Figure S3 of the Supporting Information, indicating good dielectric properties.

To evaluate the impact of O_2 PMA on the defect passivation of the IGO channel layer and its subsequent effects on device performance, BG IGO transistors with an L_{ch} of 2 μm were fabricated and characterized. The transfer characteristics of IGO transistors subjected to O_2 PMA for 3 or 10 min are shown in Figure 1c. Notably, the device annealed for 10 min has a more positive V_{th} than the device annealed for 3 min, which aligns with the theoretical expectations. This positive V_{th} shift is attributed to the passivation of interface traps at the IGO/HfO_2 dielectric interface and the suppression of oxygen vacancies within the IGO channel layer, effectively reducing the carrier concentration.[21−23] However, this process concurrently leads to degradation in the μ of the IGO channel. As shown in Figure 1d, the extracted mobilities for the 3 min- and 10 min-annealed devices are 60.2 and 38.2 $cm^2/(V{\cdot}s)$, respectively. The corresponding output characteristics (Figure 1e,f) further reveal that I_{ds} decreases from 180.1 to 140.5 $\mu A/\mu m$ after 10 min of PMA. All drain currents reported in this study were normalized to the channel width (W) to enable direct performance comparisons. These results indicate that, while prolonged O_2 PMA increases the V_{th} from −1.4 to −0.8 V, it also results in pronounced decreases in μ and I_{DS}.

To address this issue, an in situ O_3 treatment was developed to selectively passivate oxygen vacancies in the channel region, ensuring a positive V_{th}, while maintaining a high concentration of oxygen vacancies in the contact region to achieve a low R_c. A PMA of 10 min was used, as depicted in Figure 2a. Extensive O_2 passivation leads to the complete filling of oxygen vacancies in both the channel and contact regions, which consequently results in reduced carrier mobility and increased R_c. By employing the in situ O_3 passivation process, as depicted in Figure 2b, cyclic treatment involving 3 s of O_3 exposure followed by a 10 s N_2 purge was applied multiple times. This short-duration treatment leverages the strong oxidizing properties of O_3 while limiting overpassivation. Moreover, the source/drain regions, protected by the Ni/Au metallization layer, remained unaffected by O_3 treatment, preserving their high carrier concentrations. As a result, a positive V_{th} was achieved while maintaining relatively high carrier mobility in the channel layer. X-ray photoelectron spectroscopy (XPS) analysis (Figures 2c, d) demonstrated that the V_o concentration in the IGO film treated with 90 s of O_3 was reduced to approximately 10%, representing a 35% decrease compared with that of the control sample without O_3 treatment. This confirms the efficacy of O_3 in passivating V_o.

To assess the advantages of O_3 in situ treatment on the performance of practical devices, we fabricated DG transistors subjected to PMA treatment for 10 min and PMA for 3 min followed by O_3 treatment for 90 s. Compared with the BG and TG devices, the DG devices exhibited enhanced gate control, as demonstrated in Figure S4 of the Supporting Information, thereby revealing superior device characteristics. The extracted interface states density D_{it} for the top-gate structure was 7.5×10^{12} $eV^{-1}{\cdot}cm^{-2}$, which is lower than that of the back-gate configuration (1.5×10^{13} $eV^{-1}{\cdot}cm^{-2}$), demonstrating better interface quality in the top-gate architecture. The transfer and output characteristics of both configurations are presented in Figure 2e,f, respectively. As shown in Figure 2e, both devices exhibit almost identical positive V_{th}; however, the device treated with PMA for 3 min and O_3 for 90 s has an improved SS and higher I_{DS}. This suggests that O_3 treatment not only passivates V_o in the channel, leading to a positive shift in V_{th}, but also preserves the high carrier concentration in the source and drain regions, thereby mitigating the relative degradation in carrier mobility. Additionally, O_3 treatment effectively passivates interface defects at the gate dielectric/IGO channel interface, reducing the interface trap density and associated capacitance, which in turn optimizes the SS of the TFTs.[24] As shown in Figure 2f, the output current of the device treated with PMA for 3 min and O_3 for 90 s is 297.5 $\mu A/\mu m$, which exceeds the 248.1 $\mu A/\mu m$ observed for the PMA-only device (10 min), further corroborating the theoretical analysis.

To further investigate the influence of in situ O_3 treatment on the performance of IGO TFTs and optimize the O_3 treatment duration, IGO TFTs were fabricated with O_3 treatment times of 30, 60, 90, and 120 s. The transfer characteristics of the devices subjected to various O_3 treatment durations are shown in Figure 3a. Under the conditions of L_{ch} = 100 nm and V_{DS} = 0.05 V, the V_{th} values of the O_3-treated devices were −0.60, −0.10, 0.15, and 0.22 V for treatment times of 30, 60, 90, and 120 s, respectively. A clear positive correlation between V_{th} and O_3 treatment duration was observed, which can be attributed to the strong oxidative nature of O_3. This treatment effectively passivates V_o defects in the channel. As the O_3 treatment time increased, the magnitude of the positive V_{th} shifts gradually decreased, indicating a reduction in the number of passivated defects and saturation of defect passivation. The O_3 treatment led to a substantial positive V_{th} shift, while importantly, the carrier mobility remained stable without degradation. Figure 3b shows the variation in g_m with O_3 treatment time for devices with a channel length of 2 μm. As the O_3 treatment time increased, g_m exhibited a generally positive shift; however, the peak g_m value remained relatively constant at approximately 44 $\mu S/\mu m$. This suggests that, despite prolonged O_3 treatment, the mobility of the devices remained largely unaffected, maintaining a high value of 52 $cm^2/(V{\cdot}s)$, as shown in Figure 3c. Compared with the mobility of 60.19 $cm^2/(V{\cdot}s)$ observed for the PMA-treated devices with a 3 min treatment time, the mobility slightly decreased. However, it was still significantly higher than the 38.23 $cm^2/(V{\cdot}s)$ for devices treated with PMA for 10 min, highlighting that in situ O_3 treatment effectively slowed the degradation of mobility. This improvement is attributed to the preservation of a high carrier concentration in the contact

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Nano Lett. 2025, 25, 8541−8546

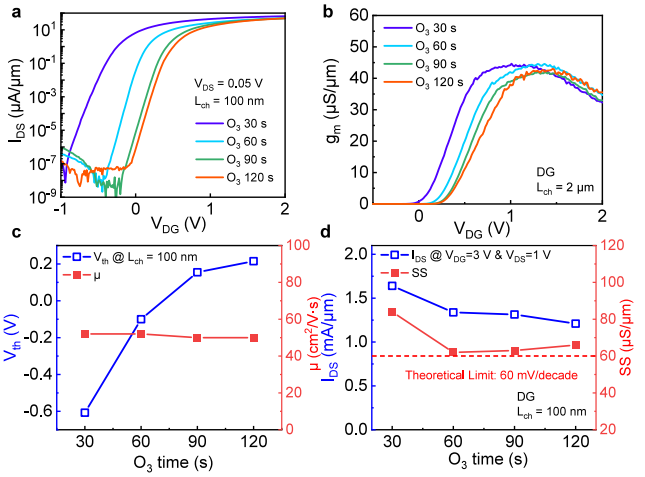


Figure 3. (a) Transfer characteristic curves for short-channel devices with L_{ch} = 100 nm, which exhibit variations with different O_3 treatment durations. As the O_3 treatment time increases, V_{th} becomes progressively more positive. (b) The g_m curves for long-channel devices with L_{ch} = 2 μm show variations with different O_3 treatment durations. As the O_3 treatment time increased, the reduction in g_m was relatively small. (c) The variations in V_{th} and μ with increasing O_3 treatment time indicate that while V_{th} shifts positively, the decrease in mobility is relatively modest. (d) Variations in I_{DS} and SS with respect to different O_3 treatment durations.

regions during the in situ O_3 treatment process. Moreover, we analyzed the I_{DS} and SS values under different O_3 treatment times, and the data are summarized in Figure 3d. As the O_3 treatment time increased, the I_{DS} gradually decreased, which was associated with the reduction in the overdrive voltage caused by the positive V_{th} shift. Moreover, the SS approached the theoretical limit at room temperature after extended treatment, indicating that O_3 treatment effectively passivates the interface defects between the gate dielectric and the IGO channel. These observations demonstrate the excellent passivation capability of the in situ O_3 treatment. A comparison of V_{th} revealed that the V_{th} for the 90 s O_3 treatment shifted almost fully to saturation, whereas the 120 s O_3 treatment resulted in a more positive V_{th}, leading to a reduced on-state current. Both the SS and I_{DS} values for the 90 s O_3 treatment were optimal, making it the most suitable treatment time in this study.

We fabricated a series of devices with L_{ch} values ranging from 100 nm to 2 μm under the optimized 90 s O_3 treatment conditions to evaluate their scaling behavior and demonstrate the performance advantages enabled by this process. The transfer and output characteristics for devices with varying channel lengths are provided in Figures S5 and S6 in the Supporting Information. Figure 4a shows the DG transfer characteristics of the L_{ch} = 100 nm device at V_{DS} = 0.05 V and V_{DS} = 1 V. Notably, V_{th} remains stable at 0.15 V even under short-channel conditions. The corresponding output characteristics, presented in Figure 4b, show a remarkable I_{DS} of 1.36 mA/μm at V_{DS} = 1 V and V_{DG} = 3 V. Moreover, as shown in Figure 4c, the device achieves an exceptionally high peak g_m of 1008 μS/μm at V_{DS} = 1 V, setting a new benchmark for IGO-based transistors. The SS remains as low as 63 mV/dec across 4 orders of magnitude in drain current, indicating excellent interface quality, as shown in Figure 4d. Figure 4e summarizes the extracted I_{DS} and g_m values across different channel lengths, which clearly exhibit inverse proportionality to channel length, confirming superior short-channel scalability. Additionally, R_c was extracted via the transfer length method (TLM), as shown in Figure 4f, revealing a minimum R_c of 0.23 kΩ·μm. The PBS values of DG/TG/BG were measured as shown in Figure S7 in the Supporting Information, indicating good bias stability. These results underscore the superior passivation and interface optimization achieved through the in situ O_3 treatment, as evidenced by the outstanding performance metrics of the optimized devices.

Figure 5a presents a benchmark plot of the I_{DS} versus SS for various oxide semiconductor devices with DG and gate-all-around (GAA) architectures,[25−33] measured under the conditions of V_{DS} = 1 V and V_{DG} - V_{th} = 1 V. The device in

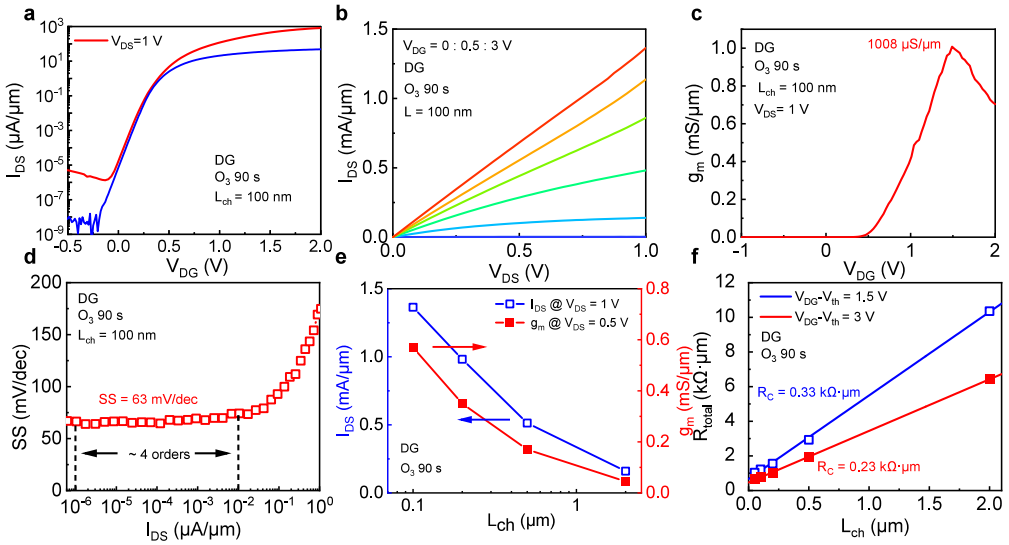


Figure 4. (a) I_{DS}-V_{DG} and (b) I_{DS}-V_{DS} of a DG IGO transistor with L_{ch} = 100 nm after 90 s O_3. (c) g_m of the DG IGO transistor with L_{ch} = 100 nm and V_{DS} = 1 V. A record high g_m of 1008 μS/μm is obtained. (d) The SS of the DG IGO transistor. An ultralow SS of 63 mV/dec is obtained, which remains at 4 orders of magnitude. (e) The I_{DS} and g_m of IGO DG-TFTs with L_{ch}. The values of g_m and I_{DS} increase with L_{ch} scaling. (f) Typical R_c of DG IGO transistors extracted by TLM (L_{ch}: 100/200/500/2000 nm) method at different V_{DG}-V_{th}.

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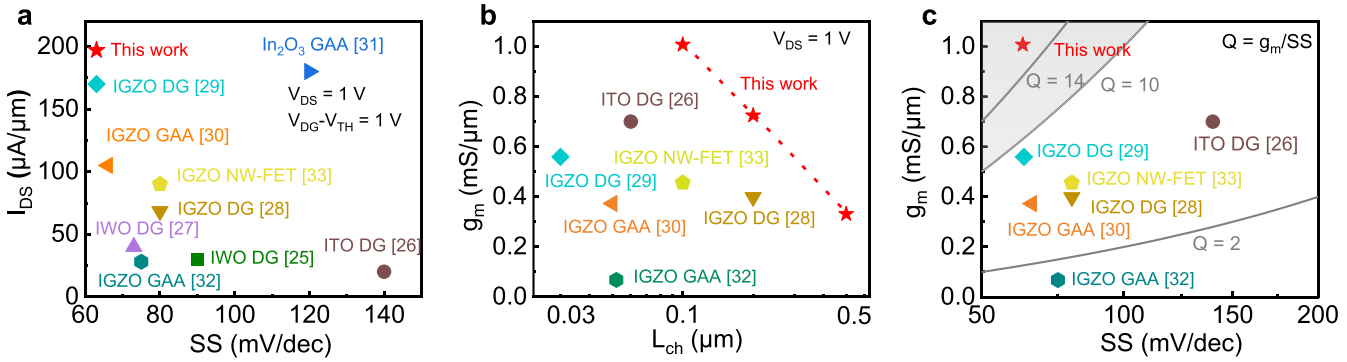


Figure 5. (a) Benchmark of I_{DS} vs SS. The I_{DS} is extracted with $V_{DS} = V_{DG} - V_{TH} = 1$ V. Our work has a high I_{DS} of 197 μA/μm and the smallest SS (b) Benchmark of g_m vs L_{ch} at $V_{DS} = 1$ V for DG, nanowire (NW), and GAA TFTs. Our work has the highest g_m of 1008 μS/μm. (c) Benchmark of g_m vs SS at $V_{DS} = 1$ V for DG, NW, and GAA TFTs. Our work has the highest quality factor of 16.5 because of the record high g_m and ultralow SS.

this work demonstrates outstanding I_{DS} and minimal SS, which can be attributed to the excellent interface quality between the gate dielectric and the IGO channel. Figure 5b compares g_m as a function of L_{ch} at $V_{DS} = 1$ V, in contrast with previously reported DG and GAA oxide semiconductor devices.[26,28−30,32,33] The IGO DG TFTs fabricated in this work exhibit an unprecedented g_m value of 1008 μS/μm, the highest value reported among all IGO DG TFTs to date. Notably, g_m increases monotonically with decreasing channel length, indicating a clear advantage for further device miniaturization and enhanced performance in scaled devices. Figure 5c shows a benchmark plot of g_m versus SS, further validating the exceptional device performance.[26,28−30,32,33] Owing to its superior interface characteristics, our device has the highest quality factor among all DG and GAA oxide semiconductor devices reported to date, highlighting its remarkable interface stability and device efficiency. This indicates the significant potential of this technology for high-performance, low-power applications. All the benchmark data above are summarized in Table 1 of the Supporting Information.

In conclusion, we have demonstrated a high-performance dual-gate IGO transistor utilizing in situ O_3 treatment. The O_3 treatment effectively passivates the defect states without compromising the carrier concentration in the contact regions. By optimizing the duration of the in situ O_3 treatment, a short-channel device with $L_{ch} = 100$ nm exhibited a maximum I_{DS} of 1.36 mA/μm, a g_m of 1008 μS/μm, and an SS of 63 mV/dec. These performance parameters represent the highest values reported to date for DG IGO transistors. This study offers a scalable and reliable methodology for achieving high-performance DG IGO transistors, presenting significant potential for the advancement of next-generation electronic devices.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.nanolett.5c01059.

Additional results and details on the fabrication process of DG IGO, the surface roughness of IGO, SEM images, and a comparison between the BG, TG, and DG configurations, transfer characteristics and output characteristics for the IGO device with 90 s of O_3 treatment, along with stability testing under positive and negative bias conditions (PDF)

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Author Contributions

C.Y. and J.L. contributed equally to this work. X.L. conceived the project. C.Y. fabricated these devices. C.Y. and J.L. performed the optical and AFM measurements. C.Y. performed the electrical characterization tests. C.Y., J.L., and X.L. analyzed the data and wrote the manuscript. All the authors commented on the manuscript.

Notes

The authors declare no competing financial interest.

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